

Route Agent

Role: Route Intelligence · Pipeline Position: Step 2 of 5 · Endpoint: POST
/route/generate

Computes Haversine distance, ETA, and fuel estimate for the active trip.

Overview

The Route Agent is the second agent in the pipeline. It validates pickup/destination coordinates, verifies the vehicle location registry, retrieves the active trip, and computes precise routing metrics including distance, estimated travel time, and fuel consumption.

Responsibilities

- Validate and parse pickup and destination coordinates.
- Verify that the assigned vehicle has a location registry entry.
- Look up the active trip linked to the vehicle.
- Calculate Haversine great-circle distance (km).
- Estimate travel time using configurable speed settings.
- Estimate fuel consumption using a configurable rate.
- Update trip record status to **Route Generated**.

API Endpoint

Method	Path	Description
POST	/route/generate	Generate route metrics for an active trip

Request Payload

```
{
  "vehicle_id": "<uuid>",
  "pickup":    "13.0827,80.2707",
  "destination": "12.9715,77.5945"
}
```

Response Payload

```
{
  "status": "success",
  "data": {
    "trip_id": "<uuid>",
    "distance_km": 347.2,
    "estimated_duration": "7h 42m",
    "estimated_fuel": 24.3
  }
}
```

Configuration Parameters

Setting	Source	Default
ROUTE_DEFAULT_SPEED_KMH	core/config.py	45.0 km/h
ROUTE_FUEL_L_PER_100KM	core/config.py	7.0 L/100km

Internal Logic Flow

```
generate_route(vehicle_id, pickup, destination)
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■■■ 1. parse_coordinates(pickup)           → InvalidCoordinateException if invalid
■■■ 2. parse_coordinates(destination)       → InvalidCoordinateException if invalid
■■■ 3. get_vehicle_location(vehicle_id)     → VehicleUnavailableException if not found
■■■ 4. get_trip(vehicle_id)                 → AgentFleetException if no active trip
■■■ 5. haversine_distance()                 → distance_km
■■■ 6. estimate_eta(distance_km, speed)     → minutes → formatted string
■■■ 7. estimate_fuel(distance_km, rate)     → fuel_liters
■■■ 8. update_trip_route(status='Route Generated')
```

Module Files

File	Purpose
agent.py	Agent registry entry — wraps service call
service.py	Business logic: coordinate validation, Haversine, fuel & ETA

repository.py	DB queries: vehicle location, active trip, route update
routes.py	FastAPI router — POST /route/generate
schemas.py	Pydantic request/response models
prompts.py	AI prompt templates
tools.py	LLM tool definitions

Error Cases

Condition	Exception	HTTP Code
Invalid pickup/destination coordinates	InvalidCoordinateException	422
Vehicle ID not in location registry	VehicleUnavailableException	503
No active trip found for vehicle	AgentFleetException	400